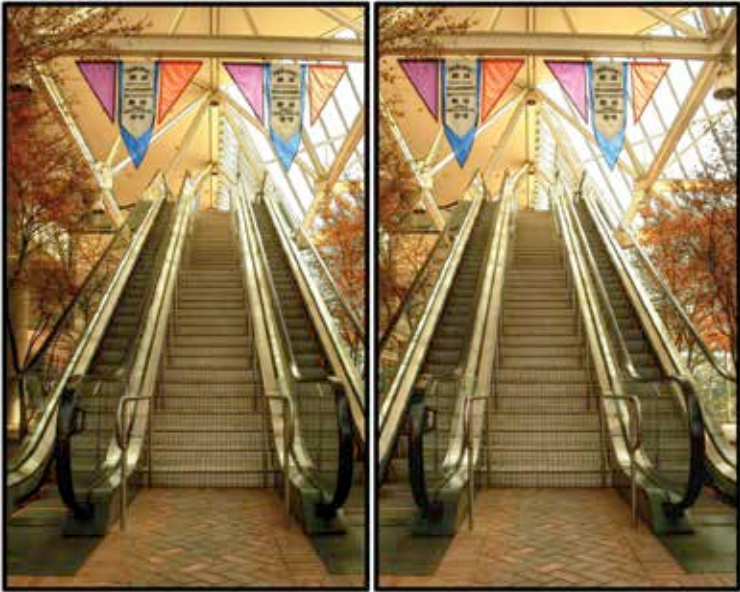




A typical stereoscopic image, where the left and right images differ slightly in perspective

otherwise flickering is noticeable to the human eye. Oculus Rift ships with a 90fps display whereas PlayStationVR supports up to 120fps.

Head Tracking

In VR, one of the main tenets of the experience is that the display should follow your head as you look anywhere around you, that too without lag. A number of VR headsets achieve this in different ways. But almost all of them have been touting 3DoF or 6DoF tracking, where DoF is Degrees of Freedom.

Any stationary object can move in 6 different ways. Think of the first three as lateral movement along the x, y and z axis, or simply, forward-backward, left-right, and up-down. The next three are rotational motions along the same axes. This might be a bit harder to visualise, but the first is leaning forward-backward, second is leaning left-right and third is turning left-right. The movement of any object, a tennis ball or a VR headset, can be defined using these six motions. For example, an elevator has 1DoF - up and down. On the rotational front, a perfect example for 1DoF is a windmill that can only rotate sideways.

One of the most popular movement detection devices is an Inertial Measurement Unit, or an IMU. These devices consist of a combination of